

Unit 11 - Ch. 15 Studyguide

Key Concepts

15.1 Water and Its Properties

- The high surface tension and low vapor pressure of water are the results of hydrogen bonding.
- The structure of ice is a regular open framework of water molecules held together by hydrogen bonds and arranged like a honeycomb.
- Molecules in liquid water are packed closer together than they are in ice.

15.2 Homogeneous Aqueous Systems

- In a solution, a solvent dissolves the solute. The solute becomes dispersed in the solvent.
- In the dissolving process, individual solute ions break away from the crystal. Solvent molecules surround the negatively and positively charged ions in a process called solvation and the ionic crystal dissolves.
- All ionic compounds are electrolytes because they dissociate into ions and thus can conduct electricity.
- The formula of a hydrate consists of the formula of the ionic salt followed by a dot and the number of water molecules associated with one formula unit of the salt.

15.3 Heterogeneous Aqueous Systems

- A suspension differs from a solution because the component particles of a suspension are much larger and do not stay suspended indefinitely.
- Colloids have particles smaller than those in suspensions and larger than those in solutions.

Vocabulary

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| ● aqueous solution | ● solvation |
| ● Brownian motion | ● solvent |
| ● colloid | ● strong electrolyte |
| ● electrolyte | ● surfactant |
| ● emulsion | ● suspension |
| ● hydrate | ● surface tension |
| ● nonelectrolyte | ● Tyndall effect |
| ● solute | ● weak electrolyte |

Key Equation

- $\text{Percent H}_2\text{O} = (\text{mass of water} / \text{mass of hydrate}) \times 100\%$